

Qatar-2b

R-0969

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_180606 · created 2026-07-29T17:12:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458275.76952 +/- 0.00064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.063
Transit depth (Rp/Rs)^2	7.9e-05 +/- 0.011 [%]
Orbital Inclination (inc)	86.25 +/- 1.77
Airmass coefficient 1 (a1)	1.19 +/- 0.51
Airmass coefficient 2 (a2)	-0.09 +/- 0.11
Scatter in the residuals of the lightcurve fit is	44.57 %
Best Comparison Star	#3 - [581, 122]
Optimal Aperture	2.88
Optimal Annulus	5.75
Transit Duration (day)	0.0622 +/- 0.0076

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.063 vs 0.16327 (deviation 2.2σ)
Duration fit vs published	0.0622 d vs 0.0767 d (deviation 1.63σ)
Mid-time O–C	0.4 min
Depth SNR	0.0
Residual scatter	44.57 %

Light curve

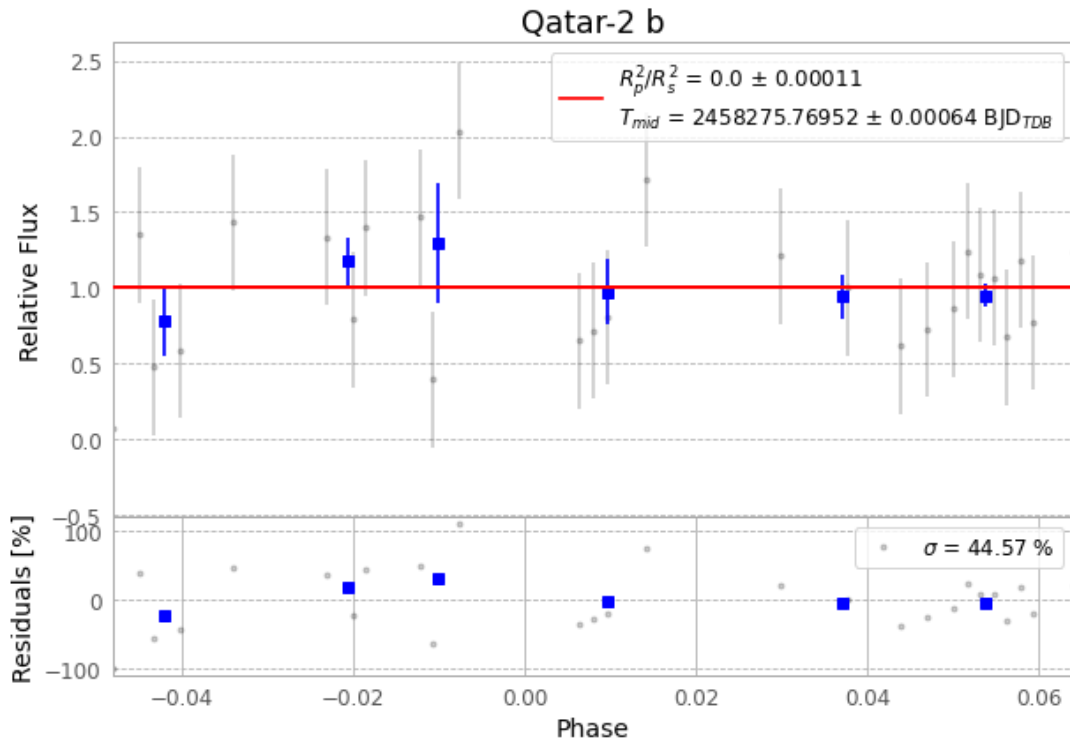


Figure 1 — FinalLightCurve_Qatar-2 b_2018-06-05.png (SHA-256 3d228a6f118fc36d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [486, 123], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 273111722558b2ef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), Qatar-2180606043912.FITS → Qatar-2180606082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-180606.log (ea457fada018...), FinalLightCurve_Qatar-2 b_2018-06-05.png (3d228a6f118f...), FinalLightCurve_Qatar-2 b_2018-06-05.csv (6f52ef1301b3...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 17:12Z.